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Unit 5 : Session 4 : SLO 2 | Architecture Fundamentals + Amdahl's Law

I. Multiple Choice

1. c) Arithmetic Logic Unit (ALU) executes the instruction's operation (the CU directs it).
2. b) RAM is volatile; loses contents without power.
3. c) Directs the flow of data and instructions.
4. a) Executing multiple instructions simultaneously in a sequence of stages.
5. b) Instruction-level parallelism (ILP).

II. True or False

1. TRUE.
2. FALSE — RAM is volatile; ROM/Flash are non-volatile.
3. TRUE.
4. TRUE.
5. FALSE — the CU controls; it does not store data permanently.

III. Scenario-Based Activity — Amdahl's Law

- Speedup with 8 cores: $S = 1 / (0.30 + 0.70/8) = 1 / (0.30 + 0.0875) = 1 / 0.3875 \approx 2.58$.
- Beyond 8 cores: no significant gains — the maximum theoretical speedup with infinitely many cores is $1/0.30 \approx 3.33$, so moving from 8 to ∞ cores gains only ~ 0.75 more; the 30% sequential portion dominates.
- Strategy for the sequential part: optimize that code itself (better algorithm, faster single-core path), or boost single-thread performance (higher clock, more cache, ILP); alternatively restructure the workload to shrink the serial fraction (e.g. parallelize parts of Task B, pipeline it with Task A).